A PADDING LOWER BOUND FOR APPROXIMATION SCHEMES FOR $P\|\sum_i C_i^q$ IN AN INTERMEDIATE-MACHINE REGIME

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Abstract. For a fixed constant $q > 1$, consider the identical-machine load-balancing problem $P\|\sum_i C_i^q$, where C_i is the load of machine i. Chen, Tao, and Verschae proved that the problem has a PTAS with running time $2^{\widetilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)}$, and that, assuming ETH, this dependence is essentially optimal for their hard instances, which have $m = \Theta((1/\varepsilon)^{1/2+o(1)})$ machines. They also asked for tight running times when $m = \Theta((1/\varepsilon)^{\theta})$ for $\theta \in (1/2, 1]$. This note records a simple padding consequence of their reduction. For every fixed $\theta \in (1/2, 1)$ and every constant $\delta > 0$, ETH rules out a PTAS in this restricted regime with running time

$$2^{O((1/\varepsilon)^{1-\theta-\delta})} + |I|^{O(1)}.$$

Consequently, for every fixed $\theta < 1$, the regime $m = \Theta((1/\varepsilon)^{\theta})$ does not admit an FPTAS unless ETH fails. The argument does not match the known $2^{\widetilde{O}(\sqrt{1/\varepsilon})}$ upper bound for $\theta > 1/2$, and it does not settle the endpoint $\theta = 1$.

1. Introduction

For $q > 1$, the problem $P\|\sum_i C_i^q$ is the following load-balancing problem. The input consists of jobs with processing times p_j and m identical machines. A feasible solution assigns each job to one machine; if C_i denotes the total processing time assigned to machine i, the objective is

$$\sum_{i=1}^{m} C_i^q.$$

Equivalently, for fixed q, this is minimization of the ℓ_q norm of the machine-load vector.

Chen, Tao, and Verschae [2] showed that, for every fixed $q > 1$, the problem admits a PTAS with running time

$$2^{\widetilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)}.$$

They also proved an essentially matching ETH lower bound: for every constant $\delta > 0$, no PTAS can run in time

$$2^{O((1/\varepsilon)^{1/2-\delta})} + n^{O(1)}$$

unless ETH fails. Their lower-bound instances have $m = \Theta((1/\varepsilon)^{1/2+o(1)})$ machines. The same paper identifies a faster polynomial-in-$1/\varepsilon$ regime when m is at least about $(1/\varepsilon)\log^2(1/\varepsilon)$, and leaves open the intermediate range

$$m = \Theta((1/\varepsilon)^{\theta}), \qquad \theta \in (1/2, 1].$$

The open-problem collection in operations research records this as Problem W4413848299_p0.[1]

This note gives a lower bound throughout the open interval $\theta \in (1/2, 1)$. The observation is that the hard instances of [2] may be padded with large dummy jobs. The dummy jobs increase the number of machines and the baseline objective value, while preserving the additive gap in

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[1]See https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4413848299_p0&n=26&page=3.

the hard core. Solving the relation between the new number of machines and the new accuracy parameter gives the exponent $1-\theta$.

Theorem 1.1. *Fix $q>1$ and $\theta\in(1/2,1)$. Assuming ETH, for every constant $\delta>0$ there is no algorithm that, on instances of $P\|\sum_i C_i^q$ satisfying $m=\Theta((1/\varepsilon)^\theta)$, computes a $(1+\varepsilon)$-approximate solution in time*

$$2^{O((1/\varepsilon)^{1-\theta-\delta})}+|I|^{O(1)}.$$

Here $|I|$ denotes the encoding length of the scheduling instance, and the constants hidden in the $O(\cdot)$ notation may depend on q,θ, and δ, but not on ε or $|I|$.

Corollary 1.2. *Fix $q>1$ and $\theta\in(1/2,1)$. Unless ETH fails, the restriction of $P\|\sum_i C_i^q$ to instances satisfying $m=\Theta((1/\varepsilon)^\theta)$ does not admit an FPTAS.*

The proof is deliberately modular. We first isolate the portion of the Chen–Tao–Verschae reduction that is used as a black box, then prove the padding lemma, and finally choose the padding size.

2. The Chen–Tao–Verschae hard core

We use the following consequence of the lower-bound construction in [2]. The parameter N below denotes the number of variables in the underlying hard 3SAT0 instance, not the number of jobs in the scheduling instance.

Proposition 2.1 (Hard core from [2])**.** *Fix $q>1$. Assuming ETH, there are constants $a_q<b_q$ and a polynomial-time reduction from a gap version of 3SAT0 with N variables to instances I_N of $P\|\sum_i C_i^q$ with the following properties.*

(i) *The number of machines is*

$$m_0=N^{1+o(1)}.$$

More precisely, in the construction of [2], $m_0=2\gamma N+8N=O(N\log N/\log\log N)$.

(ii) *There is a target load*

$$T=N^{1+o(1)}$$

such that the total processing time of all jobs in I_N is exactly m_0T. In the notation of [2], one may take $T=10^{14}\sigma_{\max}$.

(iii) *It is ETH-hard to distinguish the following two cases:*

$$\mathrm{OPT}(I_N)\le m_0T^q+a_qNT^{q-2}+o(NT^{q-2})$$

and

$$\mathrm{OPT}(I_N)\ge m_0T^q+b_qNT^{q-2}+o(NT^{q-2}).$$

More exactly, distinguishing the two cases in time $2^{O(N^{1-\rho})}$ for any constant $\rho>0$ would violate ETH.

Remark 2.2. The displayed form is a compressed version of Section 3 and Appendix G of [2]. There the common target load is $10^{14}\sigma_{\max}$, with $\sigma_{\max}=N^{1+o(1)}$, and the number of machines is $2\gamma N+8N=N^{1+o(1)}$. The yes/no objective thresholds have the same leading term $(2\gamma N+8N)(10^{14}\sigma_{\max})^q$ and differ in the coefficient of $N(10^{14}\sigma_{\max})^{q-2}$ after the 3SAT0 gap parameters are fixed.

For the rest of the note, fix one such hard instance I_N, let m_0 be its number of machines, and let T be the target load from Proposition 2.1. Let

$$\Delta_N=(b_q-a_q)NT^{q-2}+o(NT^{q-2})$$

denote the hard additive gap.

3. Padding by isolated target-load jobs

The padding construction is simple. Choose an integer $M \geq m_0$. Add $D = M - m_0$ new machines and D new dummy jobs, each of processing time exactly T. Denote the padded instance by $I_N^{\text{pad}}(M)$. It has M machines.

The key point is that the dummy jobs are forced to be alone in every optimum. Thus they add a constant term to the objective and do not alter the hard core.

Lemma 3.1 (Dummy jobs isolate)**.** *For every $M \geq m_0$,*

$$\mathrm{OPT}(I_N^{\text{pad}}(M)) = (M - m_0)T^q + \mathrm{OPT}(I_N).$$

Proof. Let the original jobs be called ordinary jobs. Their total processing time is $m_0 T$. We first show that no optimal schedule can put two dummy jobs on the same machine.

Suppose a machine has load $kT + x$, where $k \geq 2$ dummy jobs and ordinary load $x \geq 0$ are assigned to it. Since fewer than D machines contain dummy jobs, there are at least $m_0 + 1$ machines without a dummy job. At least one such machine has ordinary load $y < T$; otherwise the ordinary load on the machines without dummy jobs alone would exceed $m_0 T$. Move one dummy job from the first machine to this machine. The two affected loads change from

$$kT + x, \quad y$$

to

$$(k-1)T + x, \quad y + T.$$

The objective strictly decreases, because

$$(kT + x)^q - ((k-1)T + x)^q = \int_0^T q((k-1)T + x + s)^{q-1}\, ds.$$

and

$$(y + T)^q - y^q = \int_0^T q(y + s)^{q-1}\, ds,$$

while $(k-1)T + x > y$ and $t \mapsto t^{q-1}$ is strictly increasing. This contradicts optimality.

Therefore, in any optimum, each machine contains at most one dummy job. Suppose now that some dummy job shares its machine with ordinary load $x > 0$. Since there are exactly D dummy jobs, there are exactly m_0 machines without a dummy job. If all of those m_0 machines had load at least T, then together with the positive ordinary load x placed beside the dummy job, the ordinary load would exceed $m_0 T$. Hence some machine without a dummy job has load $y < T$. Move all ordinary jobs of total load x from the dummy machine to that machine. The two affected loads change from

$$T + x, \quad y$$

to

$$T, \quad y + x.$$

Again the objective strictly decreases, because

$$(T + x)^q - T^q = \int_0^x q(T + s)^{q-1}\, ds > \int_0^x q(y + s)^{q-1}\, ds = (y + x)^q - y^q.$$

This contradiction shows that every dummy job is alone in an optimal schedule.

After removing the D isolated dummy jobs and their machines, exactly m_0 machines remain for the ordinary jobs. Thus the ordinary part contributes $\mathrm{OPT}(I_N)$ and the dummy part contributes $DT^q = (M - m_0)T^q$. □

Remark 3.2. The proof uses only strict convexity of $x \mapsto x^q$ and the equality between the ordinary total load and $m_0 T$. It does not use any special syntactic detail of the hard instance.

4. Choosing the padding size

By Lemma 3.1 and Proposition 2.1, the padded yes/no thresholds are

$$\begin{aligned} Y_N(M) &= MT^q + a_q NT^{q-2} + o(NT^{q-2}),\\ Z_N(M) &= MT^q + b_q NT^{q-2} + o(NT^{q-2}). \end{aligned}$$

Choose a constant $\alpha > 0$ small enough that

$$0 < \alpha < \frac{b_q - a_q}{4},$$

and define

$$\varepsilon = \varepsilon(N, M) := \alpha \frac{N}{MT^2}. \tag{1}$$

Then, for N sufficiently large,

$$(1+\varepsilon)Y_N(M) < Z_N(M). \tag{2}$$

Indeed,

$$\begin{aligned} \varepsilon Y_N(M) &= \alpha \frac{N}{MT^2}\left(MT^q + a_q NT^{q-2} + o(NT^{q-2})\right)\\ &= \alpha NT^{q-2} + o(NT^{q-2}), \end{aligned}$$

where the second term is lower order because $\varepsilon \to 0$. Thus a $(1+\varepsilon)$-approximation separates the yes and no cases.

Now fix $\theta \in (1/2, 1)$ and choose

$$M = M_\theta(N) := \left\lceil \left(\frac{T^2}{N}\right)^{\theta/(1-\theta)} \right\rceil, \tag{3}$$

with the harmless convention that M is increased, if necessary, to be at least m_0. Since $T = N^{1+o(1)}$, this gives

$$M = N^{\theta/(1-\theta)+o(1)}.$$

Because $\theta > 1/2$, the exponent $\theta/(1-\theta)$ is strictly larger than 1, so $M \geq m_0 = N^{1+o(1)}$ for all sufficiently large N.

Combining (1) and (3), we obtain

$$\frac{1}{\varepsilon} = \Theta\left(\frac{MT^2}{N}\right) = \Theta\left(\left(\frac{T^2}{N}\right)^{1/(1-\theta)}\right) = N^{1/(1-\theta)+o(1)}. \tag{4}$$

Consequently,

$$M = \Theta((1/\varepsilon)^\theta) \tag{5}$$

and

$$N = (1/\varepsilon)^{1-\theta-o(1)}. \tag{6}$$

5. Proof of the main theorem

Proof of Theorem 1.1. Assume, toward a contradiction, that for some fixed $q > 1$, $\theta \in (1/2, 1)$, and $\delta > 0$, there is a $(1+\varepsilon)$-approximation algorithm for $P\|\sum_i C_i^q$ on instances satisfying $m = \Theta((1/\varepsilon)^\theta)$ with running time

$$2^{O((1/\varepsilon)^{1-\theta-\delta})} + |I|^{O(1)}.$$

Given a hard 3SAT0 instance with N variables, construct the Chen–Tao–Verschae hard core I_N and then construct the padded instance $I_N^{\mathrm{pad}}(M_\theta(N))$ with the accuracy parameter $\varepsilon = \alpha N/(M_\theta(N)T^2)$ as above. By (5), the padded instance is in the promised machine regime. By (2), the hypothetical approximation algorithm distinguishes the two cases of Proposition 2.1.

It remains to bound the running time. The padded instance has $M_\theta(N) = N^{\theta/(1-\theta)+o(1)}$ machines and only polynomially many jobs in N for fixed $\theta < 1$; the processing times have polynomial magnitude in N. Hence the encoding length $|I_N^{\mathrm{pad}}(M_\theta(N))|$ is polynomial in N. The polynomial part of the running time is therefore $N^{O(1)}$.

For the exponential part, (4) gives

$$(1/\varepsilon)^{1-\theta-\delta} = N^{1-\delta/(1-\theta)+o(1)}.$$

Since $\delta > 0$ and $\theta < 1$, this is $N^{1-\rho}$ for some constant $\rho > 0$, for all sufficiently large N. Therefore the assumed algorithm would distinguish the hard 3SAT0 gap instances in time

$$2^{O(N^{1-\rho})},$$

up to polynomial factors. This contradicts ETH via Proposition 2.1. □

Proof of Corollary 1.2. If an FPTAS existed in the regime $m = \Theta((1/\varepsilon)^\theta)$ for some fixed $\theta \in (1/2, 1)$, its running time would be bounded by $(1/\varepsilon)^K|I|^C$ for constants K, C. Choose any $\delta \in (0, 1-\theta)$. Since $(1/\varepsilon)^K = 2^{O(\log(1/\varepsilon))}$ is dominated by $2^{O((1/\varepsilon)^{1-\theta-\delta})}$, the FPTAS would contradict Theorem 1.1. □

6. What the padding argument does not prove

The lower bound in Theorem 1.1 is weaker than the known upper bound when $\theta > 1/2$. The best general PTAS of [2] runs in time

$$2^{\widetilde{O}(\sqrt{1/\varepsilon})} + n^{O(1)},$$

whereas the padding lower bound only rules out exponents below $1-\theta$. Thus the two exponents are separated whenever $\theta > 1/2$:

$$1 - \theta < \frac{1}{2}.$$

The argument also does not address the endpoint $\theta = 1$. The reason is visible from (1): after padding to M machines,

$$\frac{1}{\varepsilon} = \Theta\left(\frac{MT^2}{N}\right).$$

Since $T^2/N = N^{1+o(1)}$, the number of machines is smaller than $1/\varepsilon$ by a factor $N^{1+o(1)}$. No polynomial padding by isolated jobs can make this relation become $M = \Theta(1/\varepsilon)$ while keeping the same hard core and the same additive gap.

A matching lower bound for the whole intermediate interval would likely require a different way to increase the number of machines while also increasing the additive gap proportionally, without increasing the target load T enough to wash out the gain. One possible direction would be a product or repetition of the hard core that prevents mixing between copies on identical machines. Another would be a denser SAT-side construction whose number of consistency gadgets scales with the desired machine count. The padding argument here deliberately avoids such additional structure; it only records the consequence that follows from the existing hard core by adding isolated target-load jobs.

References

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